ECON 3420A Assignment 1: Setup and Data Guide

Software, data, the Question 5 quiz and submission

This guide covers the practical steps: installing the software, downloading your data, running the Question 5 quiz and rendering your report. The questions themselves are in `assignment_one.pdf`.

What is in the pack

File	What it is for
`assignment_one.pdf`	The questions.
`Quarto and QMD.pdf`	What a QMD file is, and how to edit and preview it in VS Code.
`solution_template.qmd`	Your answer file. Copy it to `studentID_assignment_one.qmd`.
`portfolio_model.py`	The five functions you complete with AI (Q2–Q6).
`risk_aversion_quiz.py`	The Question 5 quiz. Do not edit it.
`requirements.txt`	The Python packages to install.
`tools/download_data.py`	Downloads prices and bill yields for your own stocks.
`data/`	Hong Kong practice data. `data/practice_us/` holds a US example.

1. Install the software

Install	Why	Where
Quarto	Turns your QMD into a PDF. It includes Typst, so you do not need LaTeX.	quarto.org. Restart the terminal afterwards.
Python 3.11 or newer	Runs the calculations.	python.org. On Windows, tick "Add Python to PATH".
Python packages	Data, regressions, optimization, plots, Jupyter.	`requirements.txt`, using the commands below.

New to Quarto? Read `Quarto and QMD.pdf` first: it explains what a QMD file is and how to edit and preview one in VS Code.

From the assignment folder, create an environment and install the packages.

macOS/Linux:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
```

Windows PowerShell:

```
py -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
$env:QUARTO_PYTHON = (Resolve-Path .\.venv\Scripts\python.exe).Path
```

On Windows, use `.\.venv\Scripts\python.exe` wherever this guide says `python`. If Quarto picks the wrong Python, set `QUARTO_PYTHON` to your `.venv` Python as above.

Check that everything works:

```
python -c "import numpy, pandas, scipy, statsmodels, yfinance; print('Packages ready')"
python risk_aversion_quiz.py --help
quarto check jupyter
quarto render solution_template.qmd --to typst
```

The last command should produce `solution_template.pdf` with a practice-data table and a chart. If any step fails, ask AI to explain the **first** error message.

2. Get your data

Download

The default sample is **60 monthly returns, October 2021 to September 2026**. The downloader fetches daily prices from 1 September 2021 so that September 2021 provides the starting price.

Replace the example tickers with your own five stocks and a same-currency benchmark.

US stocks (USD):

```
python tools/download_data.py --market US \
  --stocks AAPL KO JPM XOM NVDA --benchmark SPY --out-dir data_my_stocks
```

Hong Kong stocks (HKD): keep the `.HK` suffix and leading zeros.

```
python tools/download_data.py --market HK \
  --stocks 0700.HK 0005.HK 0939.HK 0388.HK 0016.HK --benchmark 2800.HK \
  --out-dir data_my_stocks
```

Then edit the template's configuration cell:

- `prices_file`: `data_my_stocks/prices_daily.csv`
- `riskfree_file`: `data_my_stocks/riskfree_daily.csv`
- `stocks`, `benchmark` and `currency`: match `data_my_stocks/metadata.json`
- `riskfree_date_kind`: `actual quote date`

Useful options: `--start 2020-09-01 --end 2025-10-01` changes the window (first-of-month dates; the end date is excluded), and `--overwrite` replaces an earlier download.

If the HKMA bill service is down, you can use the supplied HKD bill file with your own HK stocks:

```
python tools/download_data.py --market HK \
  --stocks 0700.HK 0005.HK 0939.HK 0388.HK 0016.HK --benchmark 2800.HK \
  --riskfree-csv data/riskfree_daily.csv \
  --riskfree-date-kind source-period --riskfree-currency HKD \
  --out-dir data_my_stocks
```

In that case set `riskfree_date_kind` to `source-period label` and mention it in your data-source notes.

Sources

Input	Source	Units
Stock and benchmark prices	Yahoo Finance via yfinance	Daily close, adjusted for splits and dividends. Do not add dividends again.
USD bill rate	FRED DGS3MO	3-month Treasury yield, percent per year.
HKD bill rate	HKMA Exchange Fund Bill yields	91-day bill yield, percent per year.

Saved files

File	Contents
`prices_daily.csv`	`date`, your five stocks, then the benchmark. Missing prices stay blank.
`riskfree_daily.csv`	`date,annual_yield_percent`. `4.20` means 4.20% per year.
`metadata.json`	Tickers, currency, dates, sources and checks. Use it for your data-source notes.
`source_cache/`	The raw FRED or HKMA responses.

Practice data

`data/` holds Hong Kong practice data and `data/practice_us/` a US version, both covering the default window. Use them to test your code; they are not a substitute for your own stock choices. Their bill files contain **one quote per month**, dated with the source's period labels rather than actual quote dates. If you use them, say so.

If a download fails

Keep the error message and check the ticker, its listing date and your internet connection. Yahoo sometimes limits requests; wait and retry. A recently listed stock may not have 60 months of data: choose another stock or a documented shorter window. Never fill in missing prices or yields yourself.

3. How returns are built

The template's setup cell does this for you:

1. For each month, keep the last trading day with prices for all six assets, and compute monthly returns $R_{i,t} = P_{i,t}/P_{i,t-1} - 1$. A missing month stops the code instead of creating a two-month return.
2. For return month t, take the last bill quote in the **previous** month and convert it to a monthly rate: $R_{f,t} = q_{t-1}/1200$. For example, 4.20 becomes 0.0035.
3. Subtract it once from every stock and the benchmark: $X_{i,t} = R_{i,t} - R_{f,t}$.

For Questions 5–6, R_f is the **latest quote in your bill file** divided by 100, reported with its date.

4. Work through the report

1. Copy `solution_template.qmd` to `studentID_assignment_one.qmd` and edit its configuration cell.
2. **Questions 1–4:** complete `estimate_index_model`, `single_index_covariance` and `optimize_risky_portfolio` with AI, switch on each cell by deleting `#| eval: false`, write your answers, then render. This saves your portfolio to `outputs/q5_portfolio.json`.
3. **Question 5:** run the quiz in a terminal (next section).
4. **Questions 5–6:** complete `infer_risk_aversion` and `allocate_portfolio`, switch on the remaining cells, write your answers and render the final PDF:

```
quarto render studentID_assignment_one.qmd --to typst
```

Each render runs the whole file from the top, so every result must come from the code in the file. Keep `error: false` so a failing calculation stops the render instead of being hidden. Use `#| echo: false` to hide long code in the PDF.

5. The Question 5 quiz

Run this after rendering Questions 1–4:

```
python risk_aversion_quiz.py --portfolio outputs/q5_portfolio.json
```

It shows your portfolio's mean and volatility, the range of one-year outcomes, and five certainty equivalent wealth options A–E. No option is below your portfolio's lowest-10% outcome; if your portfolio's Sharpe ratio is high, the options are therefore closer together. Type the letter closest to your own certainty equivalent. The quiz prints your implied γ and saves your choice to `data/certainty_equivalents.csv`, which the Question 5 cell reads.

- **Just looking:** add `--show-scenarios` to see the menu without answering or saving.
- **Answering through AI chat:** AI may show you the menu, but **you choose the letter**. Save it in `data/ce_responses.csv` with two columns, `trial_id,choice`, and one row `1,C` (your letter), then run:

```
python risk_aversion_quiz.py --portfolio outputs/q5_portfolio.json \
  --responses data/ce_responses.csv
```

- **Answering again:** the quiz will not overwrite an earlier answer. Add `--overwrite`, or save elsewhere with `--output`.
- **Portfolio changed:** if you change stocks, data or the Question 4 result, re-render and take the quiz again. The Question 5 cell rejects an answer made for a different portfolio.
- **$\delta \leq 0$:** the quiz stops with a message. Skip it and explain the all-safe allocation in Question 6.

6. Submit

Open the PDF and check that every table, figure and equation appears and no placeholder is left. Submit **three files**:

- `studentID_assignment_one.qmd`
- `studentID_assignment_one.pdf`
- your completed `portfolio_model.py`